

SUPPLEMENTARY MATERIAL

CLASSIFYING INTEGRAL GROTHENDIECK RINGS UP TO RANK 5 AND BEYOND

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I. INTEGRAL DRINFELD RINGS

I.1. Up to rank 5. This section presents the comprehensive list of integral Drinfeld rings up to rank 5—including their global FPdim, type, duality, formal codegrees, and fusion data. Copy-pastable data can be found in the file `GeneralUpToRank5DataOnly.txt`, located in the `Data/General` directory of [8]. For each case, either an explicit categorification is provided, or a reference is given to a theoretical result ruling out its existence.

I.1.1. *Rank 1.* Trivial case

I.1.2. *Rank 2.*

(1) FPdim 2, type $[1, 1]$, duality $[0, 1]$, fusion data:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Formal codegrees: $[2, 2]$,
- Property: simple,
- Categorification status: $\text{Rep}(C_2)$.

I.1.3. *Rank 3.*

(1) FPdim 3, type $[1, 1, 1]$, duality $[0, 2, 1]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- Formal codegrees: $[3, 3, 3]$,
- Property: simple,
- Categorification status: $\text{Rep}(C_3)$.

(2) FPdim 6, type $[1, 1, 2]$, duality $[0, 1, 2]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

(3) FPdim 2860, type $[1, 11, 12, 12, 15, 25, 40]$, duality $[0, 1, 2, 3, 4, 5, 6]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 0 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 6 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 10 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 4 \\ 1 & 2 & 3 & 3 & 4 & 6 & 8 \\ 0 & 4 & 4 & 4 & 4 & 8 & 15 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 2 & 2 & 2 & 2 & 4 & 6 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \\ 0 & 2 & 2 & 2 & 2 & 4 & 10 \\ 0 & 4 & 4 & 4 & 4 & 8 & 15 \\ 1 & 6 & 7 & 7 & 10 & 15 & 21 \end{bmatrix}$$

- Formal codegrees: $[4, 4, 5, 5, 13, 44, 2860]$,
- Property: simple, non-1-Frobenius,
- Categorification status: open, non-braided.

(4) FPdim 3192, type $[1, 15, 16, 17, 18, 24, 39]$, duality $[0, 1, 2, 3, 4, 5, 6]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 3 & 3 \\ 0 & 0 & 1 & 1 & 1 & 3 & 3 \\ 0 & 0 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 3 & 3 & 3 & 2 & 1 & 4 \\ 0 & 3 & 3 & 3 & 4 & 4 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 3 & 3 & 3 & 4 & 4 & 8 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 1 & 1 & 2 & 2 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 3 & 2 & 4 \\ 0 & 3 & 3 & 3 & 2 & 3 & 4 \\ 0 & 3 & 3 & 3 & 4 & 4 & 9 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 1 & 1 & 1 & 3 & 2 & 4 \\ 1 & 1 & 2 & 3 & 2 & 3 & 3 \\ 0 & 2 & 2 & 2 & 3 & 2 & 6 \\ 0 & 4 & 4 & 4 & 3 & 6 & 8 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 3 & 3 & 3 & 2 & 1 & 4 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 3 & 3 & 3 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 3 & 2 & 6 \\ 1 & 1 & 2 & 3 & 2 & 7 & 7 \\ 0 & 4 & 4 & 4 & 6 & 7 & 12 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 3 & 3 & 3 & 4 & 4 & 7 \\ 0 & 3 & 3 & 3 & 4 & 4 & 8 \\ 0 & 3 & 3 & 3 & 4 & 4 & 9 \\ 0 & 3 & 3 & 3 & 4 & 4 & 9 \\ 0 & 4 & 4 & 4 & 3 & 6 & 8 \\ 0 & 4 & 4 & 4 & 6 & 7 & 12 \\ 1 & 7 & 8 & 9 & 8 & 12 & 18 \end{bmatrix}$$

- Formal codegrees: $[3, 3, 6, 8, 42, 57, 3192]$,
- Property: simple, non-1-Frobenius, non-3-positive
- Categorification status: open, non-braided, non-unitary

(5) FPdim 4284, type $[1, 17, 17, 20, 28, 35, 36]$, duality $[0, 2, 1, 3, 4, 5, 6]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 2 & 2 & 3 \\ 0 & 1 & 2 & 1 & 2 & 3 & 3 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 3 & 3 & 4 & 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 2 & 2 & 3 \\ 0 & 2 & 1 & 2 & 2 & 2 & 2 \\ 0 & 2 & 1 & 1 & 2 & 3 & 3 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 3 & 3 & 4 & 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 2 & 3 & 3 \\ 0 & 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & 1 & 3 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 6 & 4 & 4 \\ 0 & 3 & 3 & 3 & 4 & 6 & 6 \\ 0 & 3 & 3 & 4 & 4 & 6 & 6 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 3 & 3 & 3 & 4 & 6 & 6 \\ 0 & 4 & 4 & 4 & 7 & 8 & 8 \\ 0 & 4 & 4 & 4 & 8 & 8 & 8 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 3 & 3 & 3 & 4 & 6 & 6 \\ 0 & 4 & 4 & 4 & 7 & 8 & 8 \\ 1 & 5 & 5 & 6 & 8 & 10 & 10 \\ 0 & 5 & 5 & 6 & 8 & 10 & 11 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 3 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 3 & 3 & 4 & 5 & 5 \\ 0 & 3 & 3 & 4 & 4 & 6 & 6 \\ 0 & 4 & 4 & 4 & 8 & 8 & 8 \\ 0 & 5 & 5 & 6 & 8 & 10 & 11 \\ 1 & 5 & 5 & 6 & 8 & 10 & 11 \end{bmatrix}$$

- Formal codegrees: $[3, 4, 7, 7, 9, 51, 4284]$,
- Property: simple, non-1-Frobenius, non-3-positive
- Categorification status: open, non-braided, non-unitary

These fusion rings also invite questions in the spirit of [1, Question 1.13].

I.4. Ranks 8 and 9. There are 792 integral 1-Frobenius Drinfeld rings of rank 8 with $\text{FPdim} \leq 25000$, and 1292 such rings of rank 9 with $\text{FPdim} \leq 2000$. Copy-pastable data can be found in the files `1FrobR8d25000.txt` and `1FrobR9d2000.txt`, located in the `Data/General` directory of [8].

I.5. Addressing divisibility. The classification at rank r begins with the list of length r Egyptian fractions representing the possible

$$\sum_V \sum_{i=1}^{n_V^2} \frac{1}{n_V f_V} = 1,$$

where the (f_V) are the formal codegrees of the Drinfeld ring; see [1, §2.2]. Under the Drinfeld assumption, each formal codegree f_V divides f_1 . Moreover, the Drinfeld ring is commutative if and only if $n_V = 1$ for all V . In this case, we can restrict to Egyptian fractions of length r satisfying the divisibility condition $f_V \mid f_1$ for all V .

In the noncommutative setting, it may happen that $n_V f_V$ does not divide f_1 for some V . However, as verified in [1, §III], no such exceptions occur up to rank 8 (but they do appear at rank 9; see [1, §2.2]). Therefore, for ranks $r \leq 8$, it is safe to restrict to Egyptian fractions of length r that satisfy the divisibility condition.

At rank 9, [1, Lemma 9.3] shows that the complexified noncommutative Drinfeld ring must be isomorphic to either $\mathbb{C} \oplus M_2(\mathbb{C})^2$ or $\mathbb{C}^5 \oplus M_2(\mathbb{C})$. The exception corresponds to Egyptian fractions of length 5 or 7 with one or two terms having $n_V = 2$, and at least one violating the divisibility condition $n_V f_V \mid f_1$.

We verified that for $\text{FPdim} \leq 32000$, the values arising from these exceptional cases are already covered by those for the Egyptian fractions of length 9 satisfying the divisibility condition. Details of this computation can be found in the file `InvestNCRank9Except.txt` of the `Data/EgyptianFractionsDiv/Except` folder of [8].

II. MNSD DRINFELD RINGS

As established in [1, §8], the Grothendieck ring of any odd-dimensional integral fusion category over $\mathbb{C}$ is an MNSD integral Drinfeld ring ([1, Definition 8.6]). This section provides a complete classification of such rings up to rank 9.

II.1. Up to rank 5. There are four MNSD integral Drinfeld rings up to rank 5, contained in §I.1, namely the Grothendieck rings of $\text{Rep}(G)$, with $G = C_1, C_3, C_5, C_7 \rtimes C_3$.

(1) FPdim 7, type $[1, 1, 1, 1, 1, 1, 1]$, duality $[0, 6, 5, 4, 3, 2, 1]$, fusion data: see §I.3.1.
 (2) FPdim 39, type $[1, 1, 1, 3, 3, 3, 3]$, duality $[0, 2, 1, 6, 5, 4, 3]$, fusion data:

- Formal codegrees: $[3, 3, 13, 13, 13, 13, 39]$.
- Categorification status: $\text{Rep}(C_{13} \rtimes C_3)$.

- Formal codegrees: $[5, 5, 5, 5, 11, 11, 55]$,
- Categorification status: $\text{Rep}(C_{11} \rtimes C_5)$.

- Formal codegrees: $[3, 3, 7, 7, 43, 43, 903]$,
- Property: extension of $\text{ch}(C_7 \rtimes C_3)$,
- Categorification status: excluded by [1, §6.4].

- Formal codegrees: $[9, 9, 9, 9, 9, 9, 9, 9, 9]$,
- Property: pointed
- Categorification status: $\text{Vec}(C_3^2)$.

- Formal codegrees: $[9, 9, 9, 9, 9, 9, 9, 9, 9]$,
- Property: pointed,
- Categorification status: $\text{Vec}(C_9)$.

- Formal codegrees: $[3, 3, 19, 19, 19, 19, 19, 57]$,
- Categorification status: $\text{Rep}(C_{19} \rtimes C_3)$.

- Categorification status: excluded by [4, Theorem 5.1].

III.3.2. *1-Frobenius*. Here are the 20 noncommutative 1-Frobenius integral Drinfeld rings of rank 8:

(1) FPdim 8, type $[1, 1, 1, 1, 1, 1, 1, 1]$, duality $[0, 1, 2, 3, 4, 5, 7, 6]$, fusion data:

[illegible]

- Formal codegrees: $[4_2, 8, 8, 8, 8]$,
- Property: noncommutative,
- Categorification status: $\text{Vec}(D_4)$.

(2) FPdim 8, type $[1, 1, 1, 1, 1, 1, 1, 1]$, duality $[0, 1, 7, 6, 5, 4, 3, 2]$, fusion data:

[illegible]

- Formal codegrees: $[4_2, 8, 8, 8, 8]$,
- Property: noncommutative,
- Categorification status: $\text{Vec}(Q_8)$.

(3) FPdim 20, type $[1, 1, 1, 1, 2, 2, 2, 2]$, duality $[0, 1, 2, 3, 4, 5, 7, 6]$, fusion data:

[illegible]

- Formal codegrees: $[4, 4, 5_2, 20, 20]$,
- Property: noncommutative,
- Categorification status: $\mathcal{C}(F_5, 1, D_5, 1)$, $\mathcal{C}(F_5, 1, C_2, 1)$.

(4) FPdim 20, type $[1, 1, 1, 1, 2, 2, 2, 2]$, duality $[0, 1, 3, 2, 4, 5, 7, 6]$, fusion data:

[illegible]

- Formal codegrees: $[4, 4, 5_2, 20, 20]$,
- Property: noncommutative,
- Categorification status:

(5) FPdim 24, type $[1, 1, 1, 1, 1, 1, 3, 3]$, duality $[0, 1, 2, 3, 5, 4, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 8, 12, 12, 24]$,
- Property: noncommutative,
- Categorification status: excluded by [1, Lemma 5.3].

(6) FPdim 24, type $[1, 1, 1, 1, 1, 1, 3, 3]$, duality $[0, 1, 2, 3, 5, 4, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 8, 8, 24, 24]$,
- Property: noncommutative,
- Categorification status: $\mathcal{C}(S_4, 1, A_4, 1)$, $\mathcal{C}(S_4, 1, S_3, 1)$.

[illegible]

- (8) FPdim 78, type $[1, 1, 1, 1, 1, 6, 6]$, duality $[0, 1, 2, 3, 5, 4, 6, 7]$, fusion data:

[illegible]

- (9) FPdim 96, type $[1, 1, 1, 3, 4, 4, 4, 6]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- (10) FPdim 168, type $[1, 1, 1, 2, 2, 2, 3, 12]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- (11) FPdim 168, type $[1, 1, 1, 3, 3, 7, 7, 7]$, duality $[0, 2, 1, 4, 3, 5, 6, 7]$, fusion data:

[illegible]

- (12) FPdim 240, type $[1, 1, 1, 1, 1, 1, 3, 15]$, duality $[0, 1, 2, 3, 5, 4, 6, 7]$, fusion data:

[illegible]

- (13) FPdim 240, type $[1, 1, 1, 3, 6, 8, 8, 8]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 4, 15, 61, 3660]$,
- Property: noncommutative,
- Categorification status:

